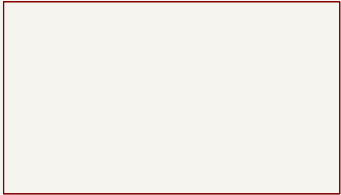


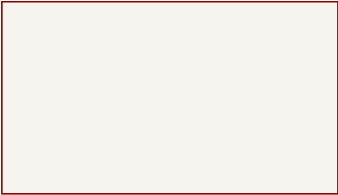
UTS Mini Mixing Desk – Compressor – one sheet per section

1 Connector



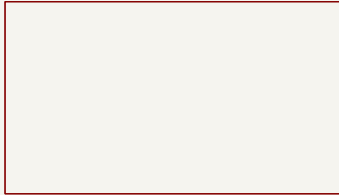
File: connector.kicad_sch

2 Input



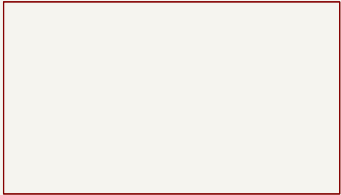
File: input.kicad_sch

3 VCA



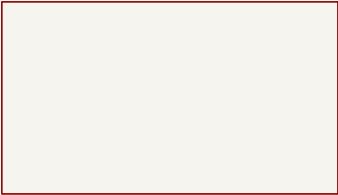
File: vca.kicad_sch

4 Output



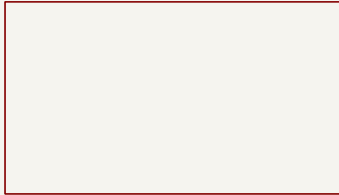
File: output.kicad_sch

5 Sidechain



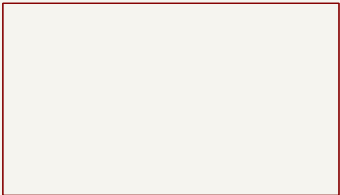
File: sidechain.kicad_sch

6 Power



File: power.kicad_sch

7 Meters



File: meters.kicad_sch

```
.param LEVEL_DBU=4 FREQ=1k
.param THRESHOLD=0.5 RATIO=1 ATTACK=0.5 RELEASE=0.5 MAKEUP=0
.param TRIM=0.5 GRTRIM=0.5 LVLTRIM=0.5
.param BYPASS=0 KEY=0 HPF_DEFEAT=0 LINK=0
* Test bench (Inspect > Simulator); see sim/README.md. Knobs run 0 to 1.
* Rack supply and ground. AGND is SPICE ground 0.
VGND AGND 0 0
VPOS +16V-IN 0 16
VNEG -16V-IN 0 -16
RPGND PGND 0 0.01
* Balanced source, 50R per leg, LEVEL_DBU differential
.param VPK={0.7746*sqrt(2)*pwr(10,LEVEL_DBU/20)}
VINP SRC+ 0 DC 0 AC 0.5 SIN(0 {VPK/2} {FREQ})
VINN SRC- 0 DC 0 AC 0.5 180 SIN(0 {-VPK/2} {FREQ})
RSP SRC+ IN+ 50
RSN SRC- IN- 50
* 20k balanced load; aux inputs and link bus open
RLP OUT+ 0 10k
RLN OUT- 0 10k
RAUXP AUXIN+ 0 1G
RAUXN AUXIN- 0 1G
RLINK LINK 0 1G
EOUT OUT_DIFF 0 OUT+ OUT- 1
ROUT OUT_DIFF 0 1G
* Solver: the 1G shunt on every node stops ngspice stalling while it compresses hard
.options method=gear rshunt=1e9
```

UTS Mini Mixing Desk

Sheet: /

File: UTS Mini Mixing Desk – Compressor.kicad_sch

Title: UTS Mini Mixing Desk – Compressor

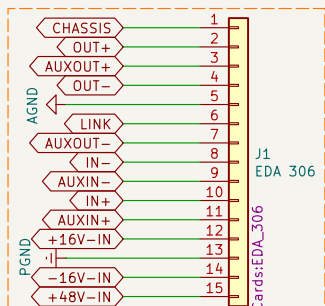
Size: A4

Date: 2026-08-31

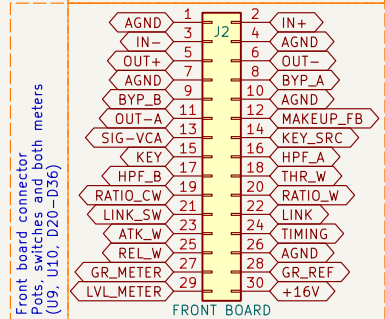
KiCad E.D.A. 10.0.6

Rev: A

Id: 1/8



On board copper female connector.
15 Poles. Labeling from Midas.



Front board connector.
Pos. switches and both meters
(U9, U10: 020-056)

UTS Mini Mixing Desk

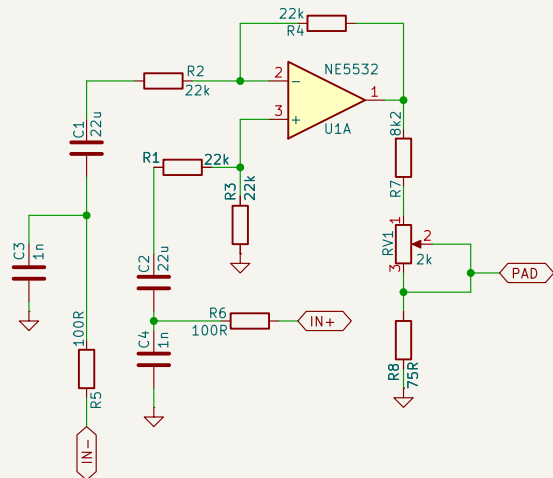
Sheet: /1 Connector/
File: connector.kicad_sch

Title: UTS Mini Mixing Desk – Compressor

Size: User Date: 2026-08-31
KiCad E.D.A. 10.0.6

Rev: A
Id: 2/8

Input Section Buffer and Unbalance the Input Before Sending it to the Pad



UTS Mini Mixing Desk

Sheet: /2 Input/

File: input.kicad_sch

Title: UTS Mini Mixing Desk – Compressor

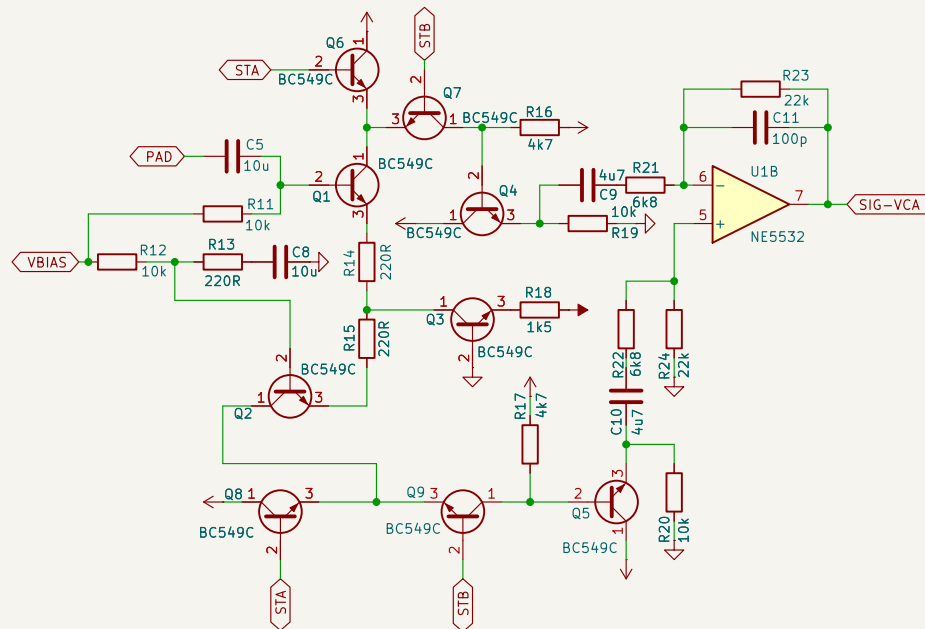
Size: User Date: 2026-08-31

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Rev: A

Id: 3/8

The Gain Cell Nine Transistors & an Op-Amp to act as a Voltage Controlled Amplifier



UTS Mini Mixing Desk

Sheet: /3 VCA/

File: vca.kicad_sch

Title: UTS Mini Mixing Desk – Compressor

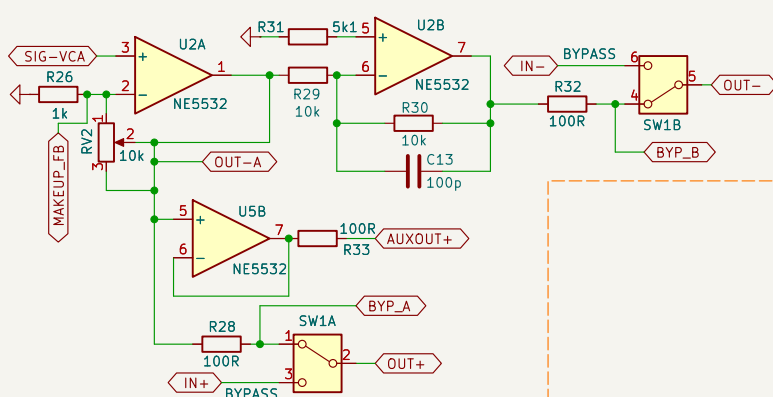
Size: User Date: 2026-08-31

KiCad E.D.A. 10.0.6

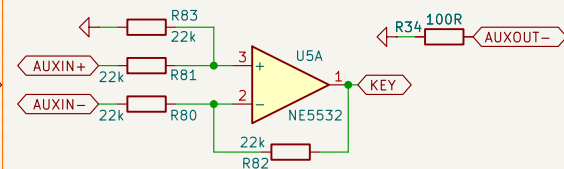
Rev: A

Id: 4/8

Output Chain, Buffer Op-Amps and Signal Balancing



Aux Output & Input Setup



UTS Mini Mixing Desk

Sheet: /4 Output/

File: output.kicad_sch

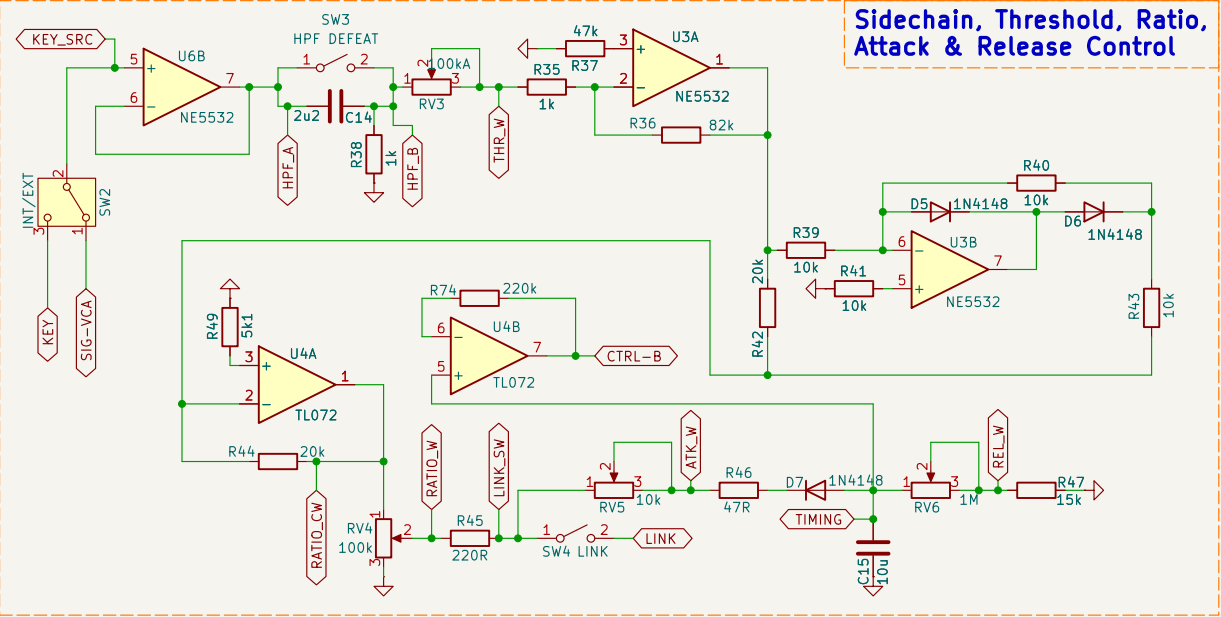
Title: UTS Mini Mixing Desk – Compressor

Size: User Date: 2026-08-31

KiCad E.D.A. 10.0.6

Rev: A

Id: 5/8



UTS Mini Mixing Desk

Sheet: /5 Sidechain/

File: sidechain.kicad_sch

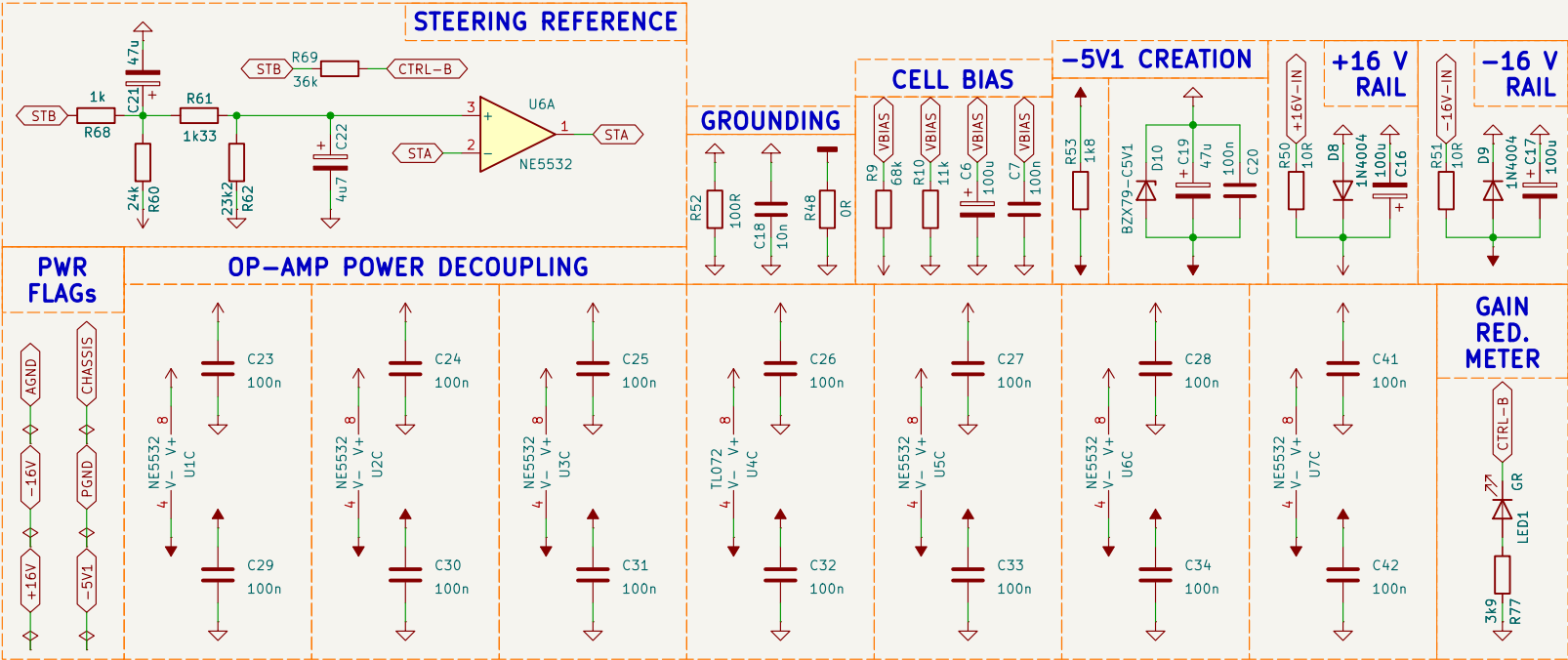
Title: UTS Mini Mixing Desk – Compressor

Size: User Date: 2026-08-31

KiCad E.D.A. 10.0.6

Rev: A

Id: 6/8



UTS Mini Mixing Desk

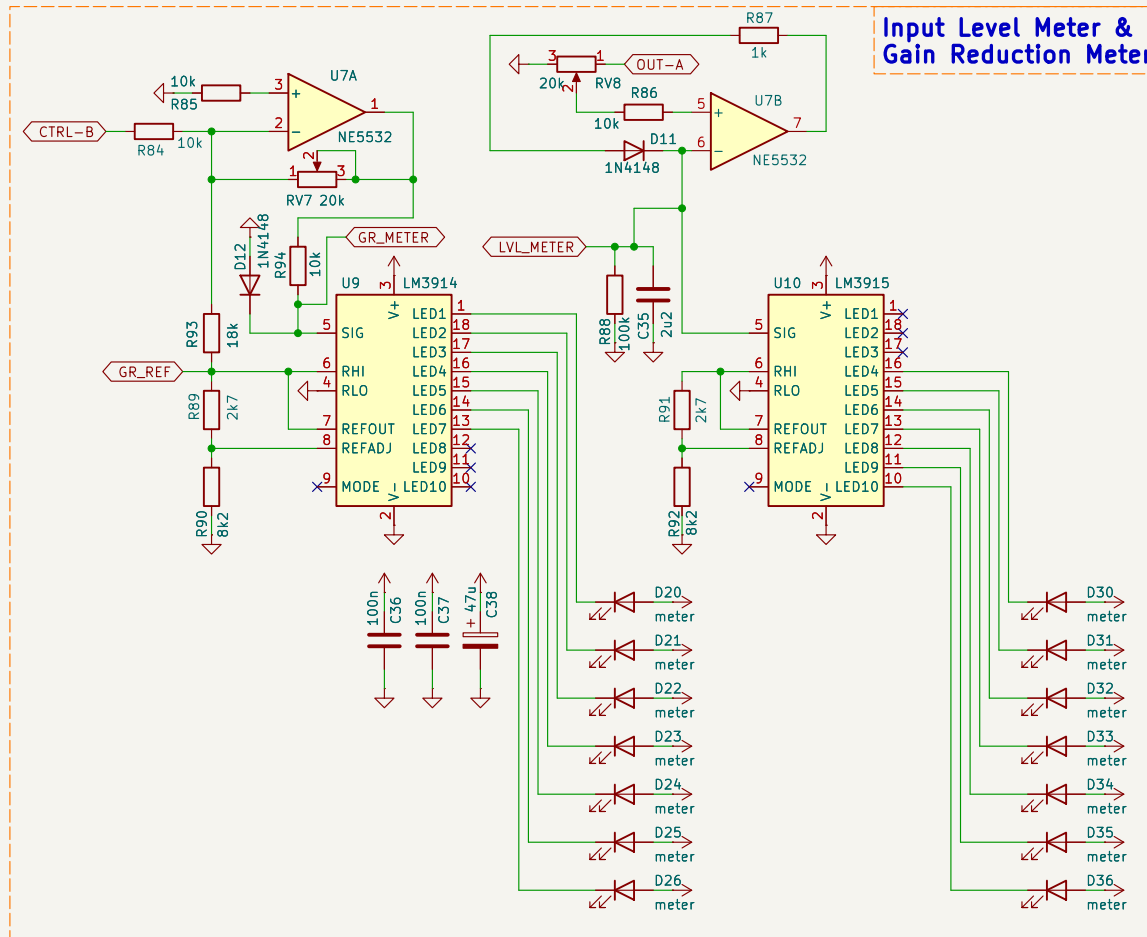
Sheet: /6 Power/
File: power.kicad_sch

Title: UTS Mini Mixing Desk – Compressor

Size: User
KiCad E.D.A. 10.0.6

Date: 2026-08-31

Rev: A
Id: 7/8



UTS Mini Mixing Desk

Sheet: /7 Meters/
File: meters.kicad_sch

Title: UTS Mini Mixing Desk – Compressor

Size: User Date: 2026-09-01
KiCad E.D.A. 10.0.6

Rev: A
Id: 8/8